

Incipient sediment motion on non-horizontal slopes

Début d'entraînement de sédiments sur des lits non horizontaux



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ABSTRACT

The study investigates the threshold condition for the initiation of cohesionless sediment transport on a non-horizontal streamwise slope. Theoretical study of the stability of a sediment particle lying on a non-horizontal bed slope shows that the critical shear stress is a function of the streamwise bed slope. Experiments conducted with a closed-conduit flow show that the equation derived from force analysis adequately evaluates the critical shear stress of sediments lying on slopes ranging from steep positive to negative (adverse). Tests conducted with a closed-conduit flow avoid the problems associated with those conducted with an open channel flow. In the latter flow condition, uniform flow is often difficult to achieve in a steep channel, and impossible to achieve when the streamwise bed slope is adverse.

RÉSUMÉ

L'étude est consacrée à la condition de début d'entraînement de sédiments non cohésifs sur une pente positive dans le sens de l'écoulement. L'étude de la stabilité théorique d'une particule sédimentaire posée sur un lit non horizontal montre que la contrainte de cisaillement critique est fonction de la pente dans le sens de l'écoulement. Des essais réalisés avec un écoulement en charge montrent que l'équation tirée d'une analyse des forces évalue la contrainte critique de cisaillement de sédiments posés sur des lits allant d'une pente positive jusqu'à une pente négative (lit en contre-pente). Ces essais réalisés avec un écoulement en charge éliminent les problèmes liés à l'écoulement en canal à surface libre. Dans un canal à surface libre, l'écoulement uniforme est souvent difficile à obtenir dans un canal à pente raide, et impossible dans un canal en contre-pente.

Introduction

The initiation of sediment motion forms an integral part of the understanding of sediment transport. The ability to quantify the threshold condition for sediment entrainment is important in the design of stable channels. The selection of the appropriate rip rap size for bank protection also requires knowledge of this threshold condition. Since the notable contribution by Shields (1936), most engineers have used some form of the Shields diagram to describe the initiation of sediment transport in cohesionless sediments. Although the Shields diagram is useful for determining incipient sediment motion, there are inherent difficulties in defining the threshold of sediment transport. The general scatter of experimental data points contained in the Shields diagram confirm this. Entrainment of sediment particles is a random process. It is a function of the fluctuating flow field, as well as the size and relative positioning of the sediment particles on the bed. The randomness of this process inevitably leads to somewhat differing definitions of the threshold condition by various researchers. Much published literature can be found on the study of the incipient motion of sedi-

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ment transport. Bibliographical references of the earlier works may be found in Vanoni (1975); Lavelle and Mofjeld (1987) have compiled a list of recently published works. Notable among the more recent contributions are the papers by Ikeda (1982) and Wiberg and Smith (1987), in which theoretical derivations of the Shields diagram are provided. In these derivations, the streamwise component of gravity acting on particles is neglected. That is, they considered the limiting case of a nearly horizontal bed, in agreement with the original data of Shields (1936).

It is clear that the critical shear stress should also be a function of the streamwise bed slope. In most open channel flows the streamwise bed slope is small, justifying the neglect of the streamwise component of the weight of sand grains. However, on a steep downward or an adverse slope, this component plays an important role in affecting the mobility of sand grains. The objective of this paper is to study the effect of streamwise bed slopes on the initiation of sediment transport of cohesionless sediments.

Incipient motion for sediment transport

Incipient sediment motion is the condition at which sediment particles just begin to move. The best known treatise on the condition for incipient motion for cohesionless sediment transport was reported by Shields (1936). He approached the problem from dimensional analysis, and proposed the following functional relationship:

$$\tau_* = F(Re_*) \quad (1)$$

and

$$\tau_* = \frac{\rho u_*^2}{(Ss - 1)dg} \quad (2)$$

where τ_* = dimensionless shear stress; F = a function; Re_* = boundary Reynolds number = u_*d/ν ; u_* = shear velocity; Ss = specific gravity of sediment; d = sediment size; g = gravitational acceleration; ρ = fluid density; and ν = kinematic viscosity.

Eq. 1 states that the dimensionless shear stress τ_* is dependent on the boundary Reynolds number Re_* . Based on flume experiments conducted using several bed materials, Shields (1936) empirically determined this functional relationship. He measured the bed load transport at different values of τ_* , which were at least two times larger than the expected critical value for sediment entrainment. By extrapolating the curves to the point of zero transport, he determined the shear stress for incipient condition for sediment motion. This indirect method for determining the incipient condition for sediment motion has its pitfalls. The most glaring of these is the determination of τ_* under conditions for which sediment transport occurs on the bed. At this stage of flow, the measured shear stress includes both shear stresses contributed by the sediment grains as well as that by the bed features. By extrapolating the shear stress to that at zero transport rate, the method may overestimate the critical shear stress.

On the other hand, Shields' method has the advantage of avoiding the precise definition of when incipient condition occurs. The precise determination of this condition in a laboratory experiment is subject to interpretation because the motion of sediment grains at low rates of transport is highly unsteady and non-uniformly distributed over the sediment bed. Observations over a large area show that the sediment motion at this state of flow to be intermittent in both time and space. This phe-

nomenon implies that the initiation of motion is statistical in nature, so that the definition of a single value depicting the condition for sediment entrainment is rather arbitrary. Despite these difficulties, variations in the results of different workers are not large, and reasonable agreement can be found in the published literature. Fig. 1 shows the Shields diagram presented by Vanoni (1975) for prediction of the critical shear stress for the initiation of sediment transport. In spite of the scatter, the data points in Fig. 1 agree reasonably well with the curve proposed by Shields (1936).

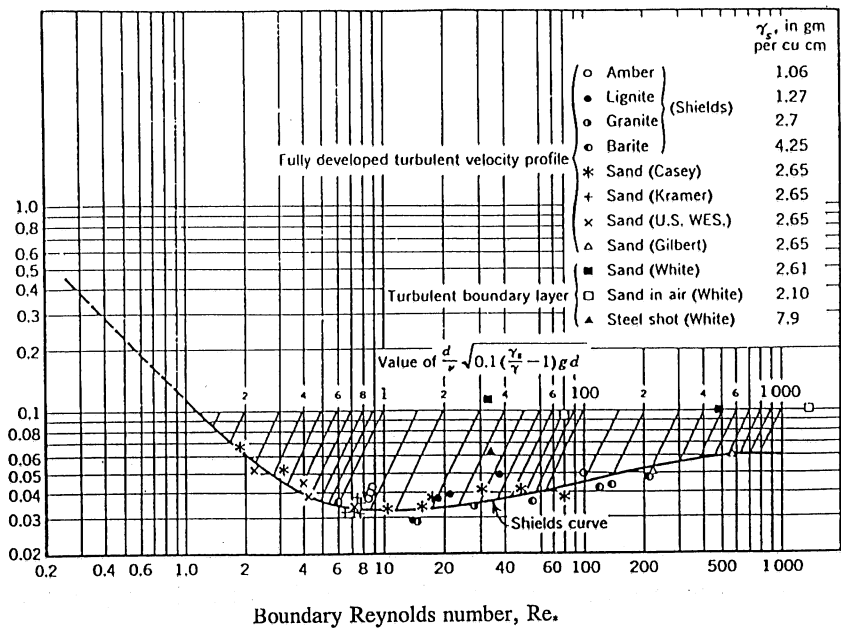


Fig. 1. Shields diagram, Vanoni (1975).

Effect of streamwise bed slope on incipient sediment motion

The effect of the streamwise bed slope on incipient sediment motion can be illustrated by considering the forces (lift, drag, buoyancy and gravity) acting on a cohesionless particle lying in a bed consisting of similar particles over which water flows (see Fig. 2). At the condition of incipient motion, the forces acting tangential (F_t) and normal (F_n) to the bed are related to the angle of repose of the sediment (θ), where

$$\tan \theta = \frac{F_t}{F_n} \tag{3}$$

The sediment particle on the bed is subject to the hydrodynamic drag force F_D , lift force F_L , and the submerged weight force W . Appropriate substitution of these forces into eq. (3) yields

$$\tan \theta = \frac{W \sin \phi + F_D}{W \cos \phi - F_L} \tag{4}$$

where $W = (\rho_s - \rho) g c_1 d^3$ is the weight of the particle, $c_1 d^3$ is the volume of the particle, ρ_s and ρ are the densities of the sediment and the fluid, respectively, and g is the acceleration due to gravity. (5)

$$F_D = \frac{1}{2} C_D \rho c_2 d^2 u_c^2 \quad (6)$$

$$F_L = \frac{1}{2} C_L \rho c_2 d^2 u_c^2 \quad (7)$$

and ϕ = angle of streamwise bed slope; and C_D , C_L = drag and lift coefficient, respectively; $c_1 d^3$, $c_2 d^2$ = volume and area of the particle, respectively; u_b = velocity of flow at the particle = $u_* \sqrt{f_*}$; f_* = a measure of friction, and u_* = shear velocity.

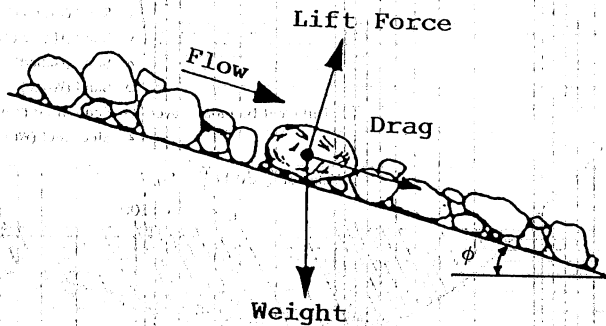


Fig. 2. Forces acting on sediment particle lying on non-horizontal slope.

By substituting the value of each of the terms, F_D , F_L and W into eq. (4) and rearranging the equation, we get

$$\tau_c = \rho u_{*c}^2 = \frac{2(\rho_s - \rho) g d c_1 f_* (\tan \theta \cos \phi - \sin \phi)}{c_2 (C_D + C_L \tan \theta)} \quad (8)$$

where τ_c and u_{*c} = critical shear stress and shear velocity, respectively.

Eq. 8 can be simplified to

$$\tau_c = K (\tan \theta \cos \phi - \sin \phi) \quad (9)$$

where

$$K = \frac{2 c_1 f_* (\rho_s - \rho) g d}{c_2 (C_D + C_L \tan \theta)} = K (Re_*) \quad (10)$$

The dependence of K on Re_* derives from the dependence of C_D , C_L , and especially f_* on the same parameter. Eq. 9 clearly shows that the critical shear stress is a function of the streamwise bed slope, ϕ for a given value of θ . For a horizontal slope, for which $\phi = 0^\circ$, eq. 9 becomes

$$\tau_0 = K (\tan \theta) \quad (11)$$

where τ_0 = critical shear stress for a horizontal bed.

Dividing eq. 9 by eq. 11, we obtain the ratio of the critical shear stress at any streamwise bed slope ϕ to the corresponding value at $\phi = 0^\circ$

$$\frac{\tau_c}{\tau_0} = \cos \phi \left(1 - \frac{\tan \phi}{\tan \theta} \right) \quad (12)$$

In terms of shear velocities, the relation takes the form

$$\frac{u_{*c}}{u_{*0}} = \sqrt{\cos \phi \left(1 - \frac{\tan \phi}{\tan \theta} \right)} \quad (13)$$

Many investigators such as Stevens et al. (1976), Luque and van Beek (1976), Smart (1984), Dyer (1986) and Whitehouse and Hardisty (1988) have used relationships similar to that of either eq. 12 or eq. 13 to determine the critical shear stress for sediment lying on a non-horizontal slope. The first of these used it to investigate the factor of safety of riprap protection, while Smart used it to evaluate sediment transport rate on a steep channel. Kovacs and Parker (1994) have recently presented a form valid for arbitrary streamwise and transverse slope that reduces to eq. 12 for a transversely horizontal bed.

There is a relative lack of experimental verification on the initiation of sediment transport on non-horizontal streamwise bed slopes. This lack may be due to the difficulties in conducting tests on steep or adverse streamwise bed slope in a laboratory flume. When the bed slope is steep, the flow becomes very difficult to control. Under this condition, supercritical flow is likely to occur and uniform flow is difficult to achieve. For an adverse slope, it is impossible to obtain uniform flow. In order to avoid the problem associated with flows on a steep or an adverse slope, the experiments in this study were conducted in a closed-conduit (pipe) flow, wherein it is possible to alter the bed slope to any desired value, including both positive (downward) and negative (adverse) angles, while still maintaining uniform flow. Here, uniform flow is defined as flow for which the velocity is invariant with distance along the channel.

Experimental setup

The experiments were conducted in two clear perspex pipes, one with a diameter of 190 mm and the other with a diameter of 50 mm. The lengths of the large and small pipe were 4 and 2 m, respectively. Fig. 3 shows the experimental set-up. A flexible joint, which acted as a hinge, connected the pipe to an upstream constant-head tank. Water that flowed out from the downstream end of the pipe was channeled into the laboratory reservoir. Three jacks were placed beneath the pipe to prevent it from bending. This set-up allowed the bed slope to be changed to any desired angle. The bed angle was varied systematically within a range of -10° (adverse slope) to $+31^\circ$ (downward slope). A submersible pump placed in the laboratory reservoir supplied water for the experiment. A gate valve in the supply line controlled the rate of flow. A flow meter was installed to provide measurement of the flow rate in the system. A second gate valve was also installed downstream of the test section to fine-tune the amount of flow in the pipe. Fig. 3 shows a cross-sectional view of the pipe and the test section. A false floor made from 1 mm thick galvanized iron sheet was placed on top of supports cut to the shape of a semi-circle. The supports were placed at regular intervals along the pipe, except for a 1 m long (0.5 m long for the small pipe) section located 2 m (1 m for the small pipe) downstream of the constant head tank. The 1 m (0.5 m for the small pipe) long section without a false floor constituted the test section wherein loose sediments were placed for testing. Sediment

particles of an appropriate size were glued on the surface of the false floor in order to simulate the roughness of the sediment bed. Altogether five sets of uniform sediments were tested in the study.

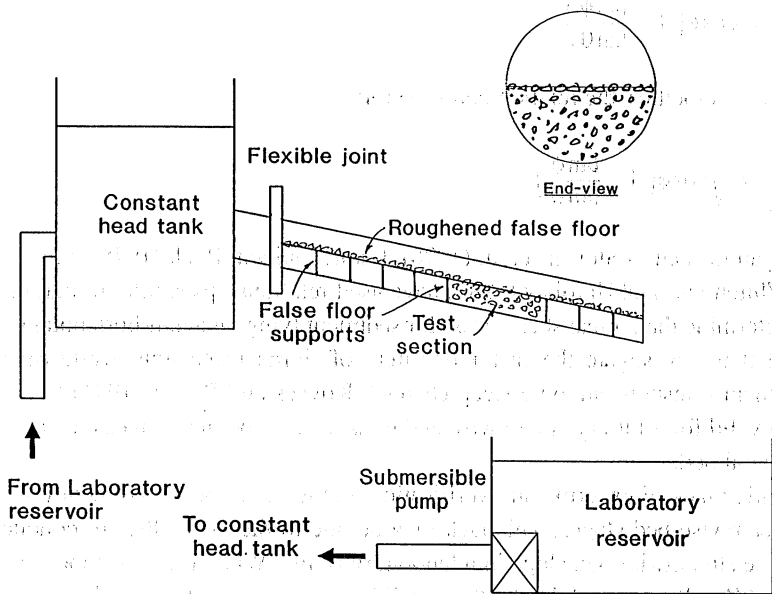


Fig. 3. Experimental set-up.

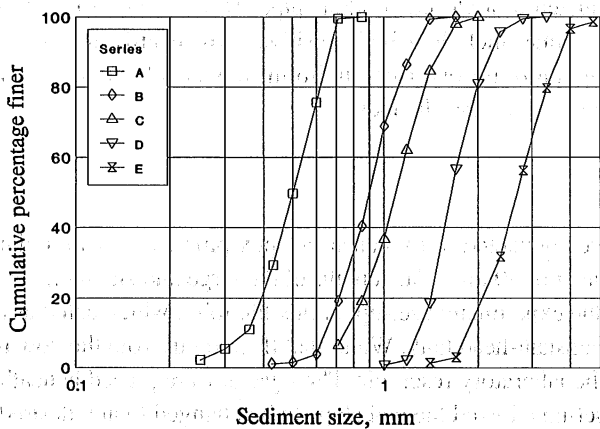


Fig. 4. Particle size distribution.

Table 1 shows the properties of the sediments used in the experiment, and Fig. 4 shows their particle size distributions. The angles of repose of the sediments were determined from the slopes of small heaps of dry sediment particles. After the sediment was placed in the test section along a predetermined streamwise bed slope, the bed was leveled so that it joined smoothly to the false floor. The pump was turned on to allow water to completely fill the pipe. The downstream gate valve was then opened slowly and observation was made on the mobility of the sand grains. The flow rate was increased by slowly opening the upstream gate valve until weak movement of sediment particles was observed in the pipe. This state

of weak movement was used to define the critical condition. Because of the method used to define critical condition for sediment entrainment, a certain amount of subjectivity and judgment is inevitable. Therefore, a corresponding amount of scatter in the experiment results is expected.

Table 1 Properties of sediments used in experiments,

median particle size, d_{50} , mm	u_{*s} m/s	u_{*0} m/s	angle of repose, θ (°)
0.50	0.0158	0.0200	33
0.91	0.0222	0.0290	34
1.07	0.0246	0.0295	36.5
1.70	0.0331	0.0410	37
2.70	0.0447	0.0820	38

Once the incipient condition for sediment motion was reached, the flow rate through the test section was recorded. The mean bed shear stress acting on the pipe diameter, τ was computed using the relation

$$\tau = \frac{\rho f V_o^2}{8} \tag{14}$$

where V_o = mean flow velocity; and f = Darcy Weisbach friction factor. The mean flow velocity, V_o can be easily computed once the flow rate is known. The friction factor f can be estimated using the standard relationship (Streeter, 1971) given below

$$\frac{1}{\sqrt{f}} = -0.861n \left(\frac{\epsilon/4r_h}{3.7} + \frac{2.51}{Re\sqrt{f}} \right) \tag{15}$$

where ϵ = bed roughness, assumed to be equal to the median grain size d_{50} ; r_h = hydraulic radius; Re = Reynolds number; and f = friction factor. Note that the length dimension used in the Reynolds number computation is four times the hydraulic radius. Eq. 15 covers the region of flow between the hydraulic smooth and completely rough zone. Once the friction factor is known, the bed shear stress can then be calculated, and the shear velocity evaluated from the definition

$$u_* = \sqrt{\frac{\tau}{\rho}} \tag{16}$$

A problem in applying eq. 14 to compute the boundary shear stress, τ is the differing nature of the bed and the side walls, the former being rough and the latter smooth. This problem is treated below.

Method for side wall correction

In many flumes used to study open channel sediment transport the bed is rough, consisting of granules, but the walls are smooth and non-erodible. As a result the Darcy-Weisbach friction factor f_w associated with the walls is significantly less than the value f_b associated with the bed. This phenomenon gives rise to a wall shear stress τ_w that is often significantly less than the bed shear stress τ_b . The method used to resolve boundary shear stress into its two components is known as the “side wall correction”. Vanoni (1975) has described such a procedure for a rectangular open-channel flume.

An analogous situation arises in the present case. The pipe perimeter consists of a smooth arc of length P_w and a horizontal granular surface P_b such that the total wetted perimeter $P = P_b + P_w$. Friction factor and bed shear stress must then be divided into the respective bed and wall components (f_b, τ_b) and (f_w, τ_w). The appropriate shear stress to be used in testing eqs. 12 or 13 is τ_b . Here the cross-sectional area A of the flow is divided into bed and wall components, one with area A_b and the other with area A_w , such that

$$A_w V_w + A_b V_b = Q = AV \quad (17)$$

The cross-sectional averaged flow velocity V is taken to be the same in the bed and wall regions. Streamwise force balance for an equilibrium flow requires that

$$-A \frac{dp}{dx} = \rho \frac{f}{8} V^2 P = \rho \frac{f_w}{8} V^2 P_w + \rho \frac{f_b}{8} V^2 P_b \quad (18)$$

where p denotes the piezometric pressure and f is an average friction factor for the entire cross-section. Eq. 18 then specifies f as

$$Pf = P_b f_b + P_w f_w \quad (19)$$

Requiring further that the piezometric pressure gradient be identical for the bed and wall regions, a force balance applied to each region yields

$$\frac{Pf}{A} = \frac{P_b f_b}{A_b} = \frac{P_w f_w}{A_w} \quad (20)$$

Eq. 20 and the definitions for total, bed region and wall region Reynolds numbers, i.e.

$$Re = \frac{4VA}{\nu P}; Re_b = \frac{4VA_b}{\nu P_b}; Re_w = \frac{4VA_w}{\nu P_w} \quad (21)$$

yield the results

$$\frac{Re}{f} = \frac{Re_b}{f_b} = \frac{Re_w}{f_w} \quad (22)$$

Insofar as the wall is taken to be smooth here, the Blasius relation (Streeter, 1971) can be used to evaluate f_w ;

$$f_w = 0.316 Re_w^{-0.25} \quad (23)$$

This equation is valid for a hydraulically smooth flow with Reynolds numbers up to 100,000. Since Re_w varies between 3,000 to 27,500 in the case of the experiments reported here, the use of eq. 23 is valid.

The parameters V, P_b, P_w and ν must be specified for any given case. The five unknowns f, f_b, f_w, A_b and A_w can then be computed with the aid of the five equations embodied in eqs. 17, 19, 22 and 23. The value of τ_b is computed from eq. 14 applied to the bed region. The calculation can be simplified by rearranging eq. 23 with the aid of eq. 22 so as to yield

$$f_w = \left(\frac{0.398}{Re} \right)^{1/5} \quad (24)$$

Incipient sediment motion on non-horizontal slopes

Table 2 shows the range of all the experimental data collected in both the large and small pipes (the detailed data can be found in Chiew and Parker, 1994). The data show that the critical shear stress for sediments lying on an adverse or a steep downward slope deviates from that lying on a horizontal or near-horizontal slope. Fig. 5 shows the ratio u_{*c}/u_{*0} (where u_{*c} and u_{*0} = critical shear velocity at any bed angle and critical shear velocity at a horizontal bed, respectively) as a function of the dimensionless ratio, ϕ/θ . Superimposed on the experimental data is eq. 13, where the angle of repose θ is taken as 36° , i.e., the average value of the five sediments used in the study (see Table 1). Despite the general scatter of the data, a trend clearly emerges showing that the experimental data support eq. 13. It must be pointed out that the experimental data deviate from eq. 13 for adverse bed slopes, with the data plotting somewhat higher than that predicted using eq. 13.

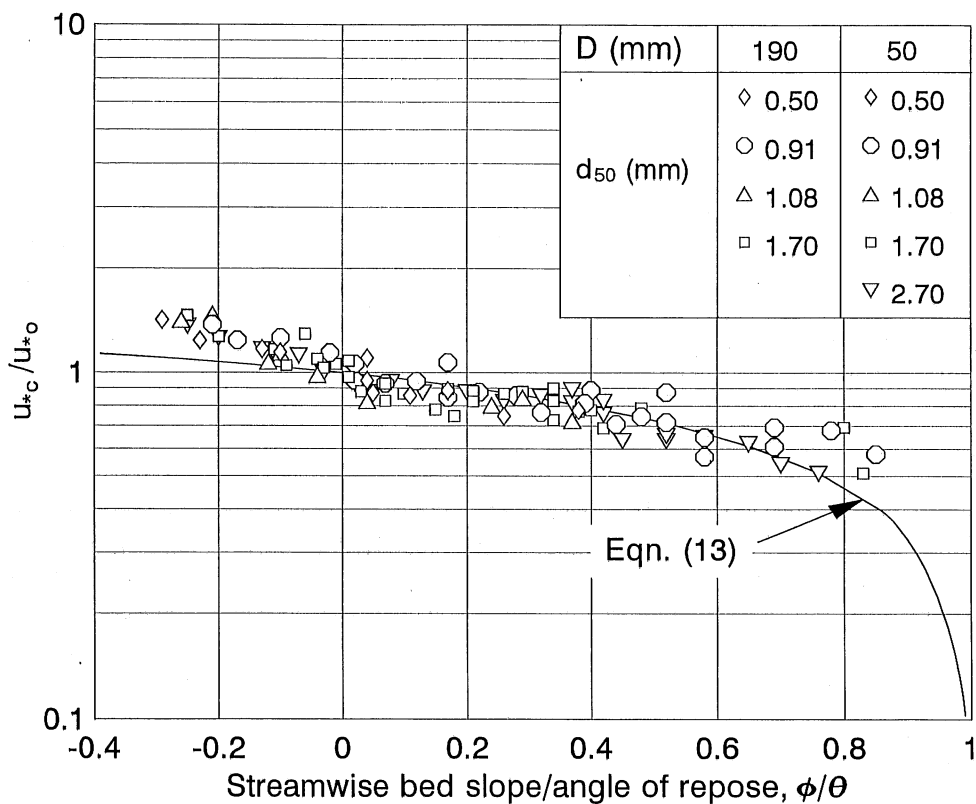


Fig. 5. Effect of streamwise bed slope on critical shear velocity.

If the Shields diagram can be accepted as the means to evaluate critical shear velocity for sediment entrainment on slopes equal or near to horizontal, u_{*0} must by necessity be equal to the critical shear velocity computed using the Shields diagram, u_{*s} . Table 1 shows both the values of u_{*0} and u_{*s} for the five uniform sediments tested in this study. It shows that the measured data are higher than that predicted using the Shields diagram. For the four sizes 0.50, 0.91, 1.07 and 1.70 mm, the measured value u_{*0} is typically 25% higher than u_{*s} . This deviation is probably within the range that might be expected due to systematically different subjective criteria for the threshold motion. In the case of

the 2.70 mm sediment, u_{*0} is 83% higher than u_{*s} . The reason for this strong discrepancy is not known. The values of u_{*c}/u_{*0} for 2.70 mm data, however, plot consistently with the data for the other sediments in Fig. 5. This fact suggests that the experimentally determined condition for the threshold motion for the 2.70 mm material is applied the same way for all the bed slopes.

Table 2 Range of experimental data collected.

Pipe diameter, mm	ϕ ($^{\circ}$)	Vel (m/s)	u_{*c} (m/s)
190	-4.46 to 17.66	0.2 to 0.516	0.0149 to 0.0531
50	-9.61 to 30.6	0.149 to 0.709	0.0165 to 0.1119

Conclusions

The study confirms that the streamwise bed slope has an important influence on the threshold condition for sediment entrainment. The critical flow condition was established analytically and experimentally. The former was achieved by analyzing forces acting on a sediment particle resting on a non-horizontal streamwise bed slope. The experiments were conducted under a closed-conduit flow. This approach successfully eliminates the difficulties of conducting experiments in steep or adverse slopes in open channel flows. Allowing for some scatter, the experimental data collapse into a unique curve when u_{*c}/u_{*0} is plotted as a function of ϕ/θ , showing the effect of the streamwise bed slope on the threshold condition for sediment entrainment. The curve of u_{*c}/u_{*0} versus ϕ/θ shows that eq. 13 can adequately describe the incipient condition for the transport of sediments lying on streamwise slopes ranging from steep downward to zero. For adverse slopes, eq. 13 underestimates the critical shear velocity.

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Notations

A	cross sectional area of flow
A_b	cross sectional area of flow on the bed
A_w	cross sectional area of flow on the wall
c_1	numerical constant;
c_2	numerical constant;
C_D	drag coefficient;
C_L	lift coefficient;
d	sediment grain size;
d_{50}	median particle size;
f	Darcy Weisbach friction factor;
f_*	a measure of friction factor;
f_b	Darcy Weisbach friction factor on the bed;
f_w	Darcy Weisbach friction factor on the wall;
F	a function of;
F_n	force acting normal to bed slope;
F_t	force acting tangential to the bed slope;
F_D	drag force;
F_L	lift force;
g	gravitational acceleration = 9.81 m/s ² ;
K	numerical constant;
P	wetted perimeter;
P_b	wetted perimeter of bed;
P_w	wetted perimeter of wall
r_h	hydraulic mean radius;
r_{hb}	hydraulic mean radius of bed;
r_{hw}	hydraulic mean radius of wall;
Re_*	boundary Reynolds number;
Re_b	Reynolds number of bed;
Re_w	Reynolds number of wall;
S_s	specific gravity;
u_*	shear velocity;
u_{*c}	critical shear velocity at any slope;

u_{*0}	critical shear velocity for horizontal slope;
u_{*s}	critical shear velocity computed using Shields diagram;
u_b	flow velocity acting on a particle;
V_b	velocity acting on bed;
V_o	mean velocity;
V_w	velocity acting on wall;
W	submerged weight force;
ε	bed roughness;
θ	angle of repose of sediment;
ν	kinematic viscosity;
ϕ	angle of streamwise bed slope;
ρ	density of water = 1000 kg/m ³ ;
ρ_s	density of sediment = 2650 kg/m ³ ;
τ_*	dimensionless shear stress; and
τ_c	critical shear stress; and
τ_o	critical shear stress for horizontal slope.